

DeePM-GAT K=10 Technical Audit

Data: Yahoo public proxy snapshot data_20260625.parquet.

Model: deepm-gat / DeePM; evaluation path: DeepMomentum seed ensemble.

Ensemble: top 10 validation-ranked seed checkpoints per rolling test window.

Windows: 2010-2014, 2015-2019, and 2020 through available 2026 data.

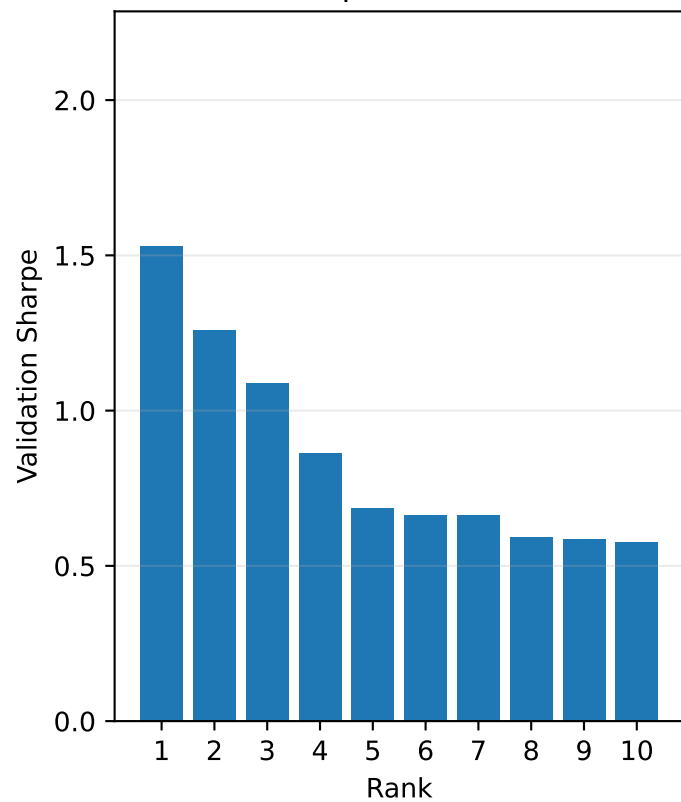
Caveat: These are diagnostic Yahoo proxy results, not paper-equivalent proprietary futures results.

Training Artifact Selection Summary

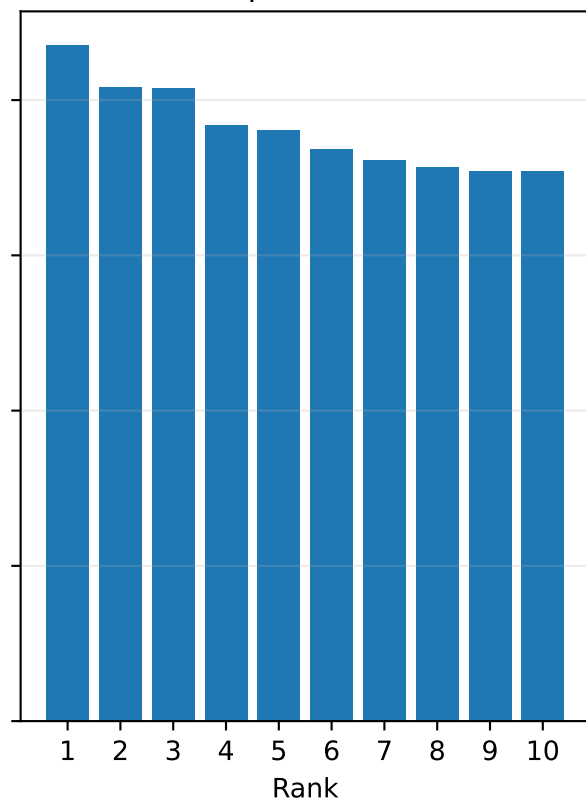
test_start	usable_runs	selected_runs	best_valid_sharpe	mean_selected_valid_sharpe
2010	25	10	1.528	0.8505
2015	21	10	2.177	1.905
2020	34	10	1.927	1.648

Selected Seed Validation Ranking

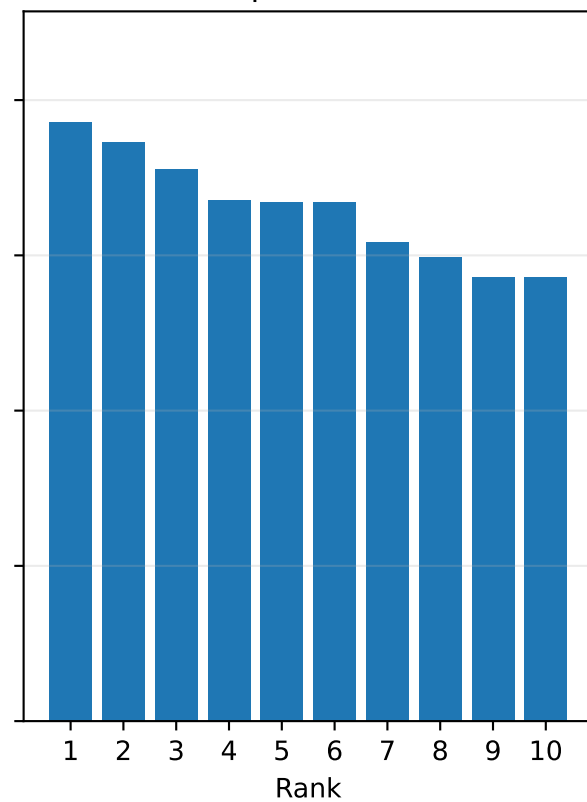
2010 top 10 validation



2015 top 10 validation



2020 top 10 validation



Selected Seed Metrics

test_start	rank	run_name	valid_loss_best	test_sharpe_gross	test_sharpe_net	test_calmar_gross	test_calmar_net
2010	1	stilted-sweep-50	1.528	0.8854	0.5309	3.148	1.894
2010	2	fancy-sweep-46	1.259	0.825	0.3777	3.252	1.494
2010	3	playful-sweep-23	1.089	0.6077	0.3327	3.054	1.673
2010	4	frosty-sweep-47	0.8635	1.082	0.7356	3.892	2.676
2010	5	stilted-sweep-28	0.6866	0.9384	0.8043	6.768	5.804
2010	6	glamorous-sweep-4	0.6637	1.088	0.7248	4.672	3.114
2010	7	lemon-sweep-12	0.6637	0.7169	0.5498	4.966	3.809
2010	8	youthful-sweep-24	0.5924	0.6701	0.3757	4.138	2.322
2010	9	fallen-sweep-5	0.5839	0.5654	0.5107	4.778	4.317
2010	10	glad-sweep-32	0.5752	0.6237	0.5013	4.912	3.95
2015	1	balmy-sweep-11	2.177	2.077	1.608	9.118	7.493
2015	2	different-sweep-22	2.04	1.668	1.182	6.642	4.709
2015	3	peach-sweep-47	2.039	1.518	0.9862	2.633	3.026
2015	4	confused-sweep-21	1.92	2.091	1.499	4.136	5.723
2015	5	hopeful-sweep-44	1.901	1.514	0.9052	4.782	2.859
2015	6	visionary-sweep-37	1.84	1.881	1.387	5.952	4.385
2015	7	splendid-sweep-6	1.806	1.281	0.751	6.194	3.634
2015	8	lilac-sweep-45	1.783	2.041	1.47	6.035	4.343
2015	9	exalted-sweep-28	1.772	2.042	1.551	8.924	6.78
2015	10	vocal-sweep-52	1.772	1.778	1.241	6.166	4.304
2020	1	honest-sweep-2	1.927	1.606	1.13	0.4929	0.8478
2020	2	elated-sweep-68	1.862	0.8251	0.4469	1.222	1.012
2020	3	jumping-sweep-61	1.778	0.864	0.4801	3.855	2.153
2020	4	lucky-sweep-58	1.676	0.8531	0.4867	3.77	2.157

Main Metrics Comparison

Model	Sharpe	Calmar	CAGR%	Vol%	MaxDD%	t-stat	Hit%
Long Only	0.57	0.1802	5.336	10	-29.61	2.351	53.98
TSMOM	0.4264	0.1586	3.835	10	-24.17	1.759	52.6
Risk-Managed TSMO	0.4179	0.1393	3.746	10	-26.88	1.724	52.44
MVO Trend	-0.06571	-0.02355	-1.151	10	-48.87	-0.271	50.27
MVO-TP Trend	0.3994	0.1406	3.555	10	-25.29	1.648	52.79
ERC Trend	0.3002	0.08403	2.533	10	-30.15	1.238	51.55
MACD	-0.08733	-0.03176	-1.364	10	-42.97	-0.3602	52.2
Risk-Managed MACD	-0.0742	-0.0297	-1.235	10	-41.58	-0.306	52.09
MVO-TP MACD	-0.1086	-0.03298	-1.575	10	-47.75	-0.4481	51.78
PM-GAT (K=10 Yahoo	1.093	0.5309	10.99	10	-20.71	4.509	53.79

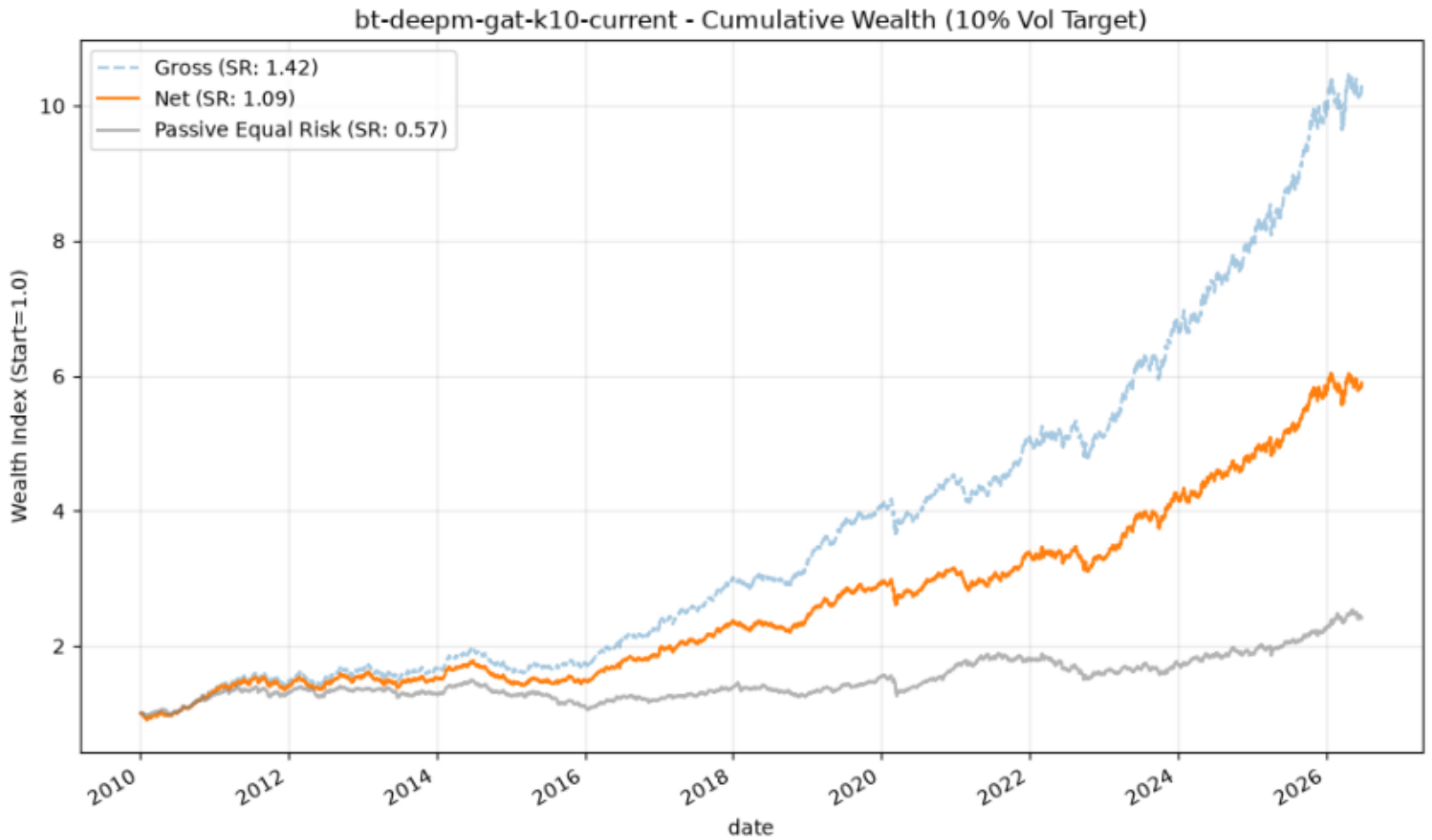
DeePM Net Metrics

metric	value
CAGR	0.1099
Annualized Return	0.1093
Annualized Vol	0.1
Sharpe Ratio	1.093
t-statistic	4.509
t-statistic (HAC)	4.389
Worst Window Sharpe (3m)	-5.361
Minimum Annual Sharpe	-0.415
Calmar Ratio	0.5309
Max Drawdown	-0.2071
CVaR 5% (Loss)	0.01385
Hit Rate	0.5379
Annualized Turnover (Scaled)	328.6
Turnover Ratio (xGMV)	46.59
Average Leverage (Scaled)	7.052
Avg Holding Period (Days)	10.82
Breakeven Cost (bps)	3.327
t-statistic vs Passive (HAC)	2.304
Information Ratio	0.5444
Correlation vs Passive	0.538
Ex-COVID Gross Sharpe Ratio	1.453
Ex-COVID CAGR	0.1144
Ex-COVID Annualized Return	0.1134
Ex-COVID Annualized Vol	0.1

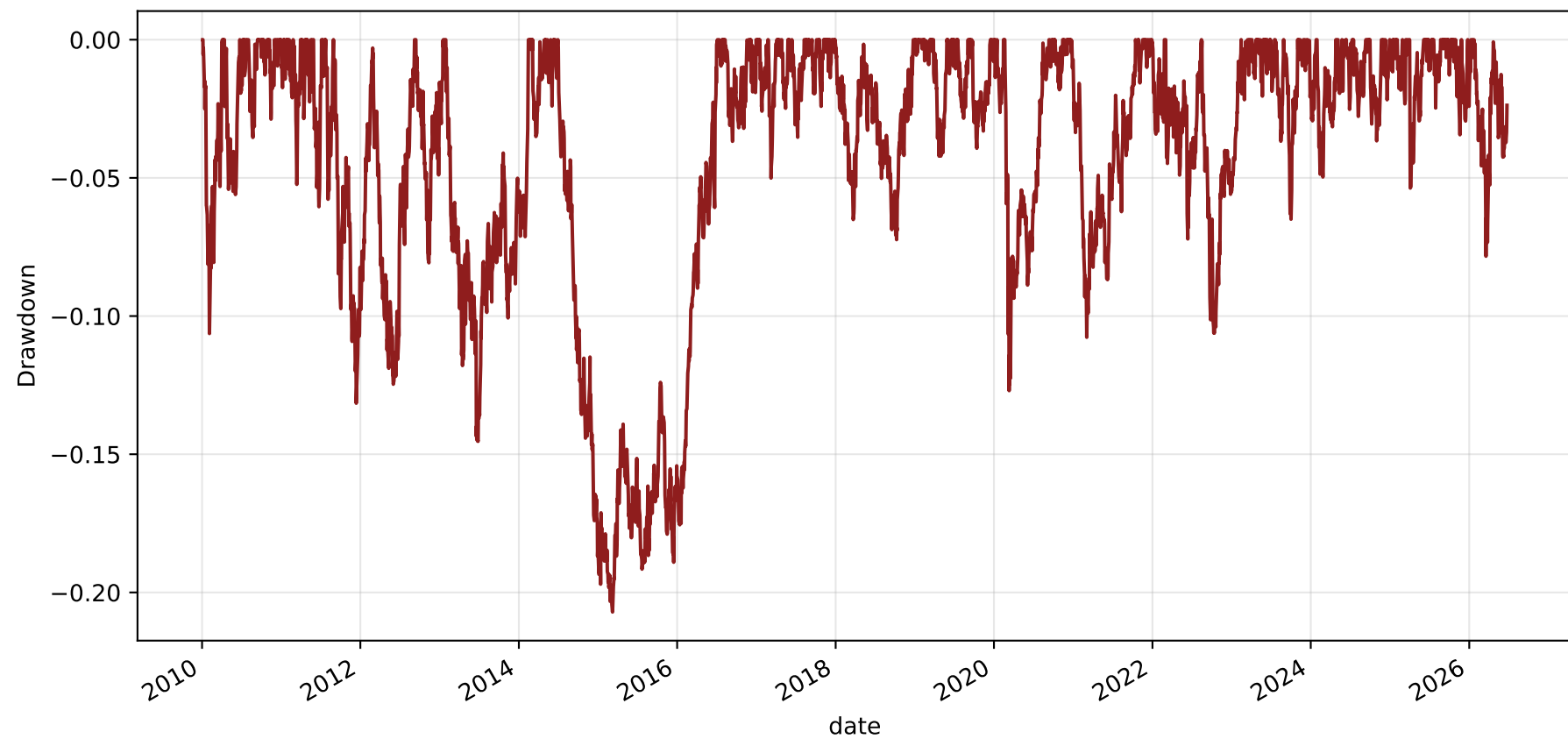
DeePM Gross Metrics

metric	value
CAGR	0.147
Annualized Return	0.1422
Annualized Vol	0.1
Sharpe Ratio	1.422
t-statistic	5.863
t-statistic (HAC)	5.713
Worst Window Sharpe (3m)	-4.994
Minimum Annual Sharpe	-0.1393
Calmar Ratio	0.7837
Max Drawdown	-0.1875
CVaR 5% (Loss)	0.01373
Hit Rate	0.5477

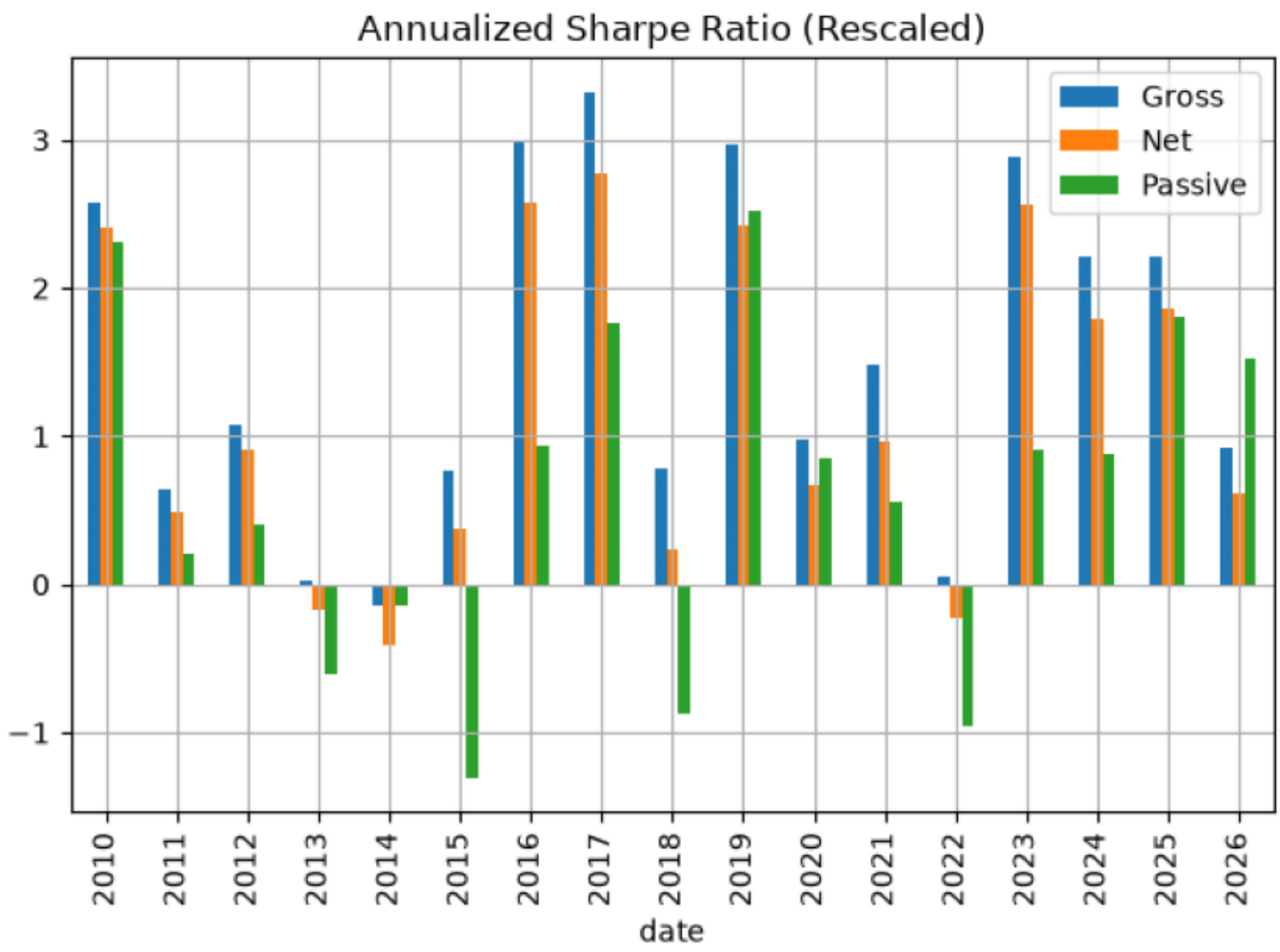
Cumulative Wealth



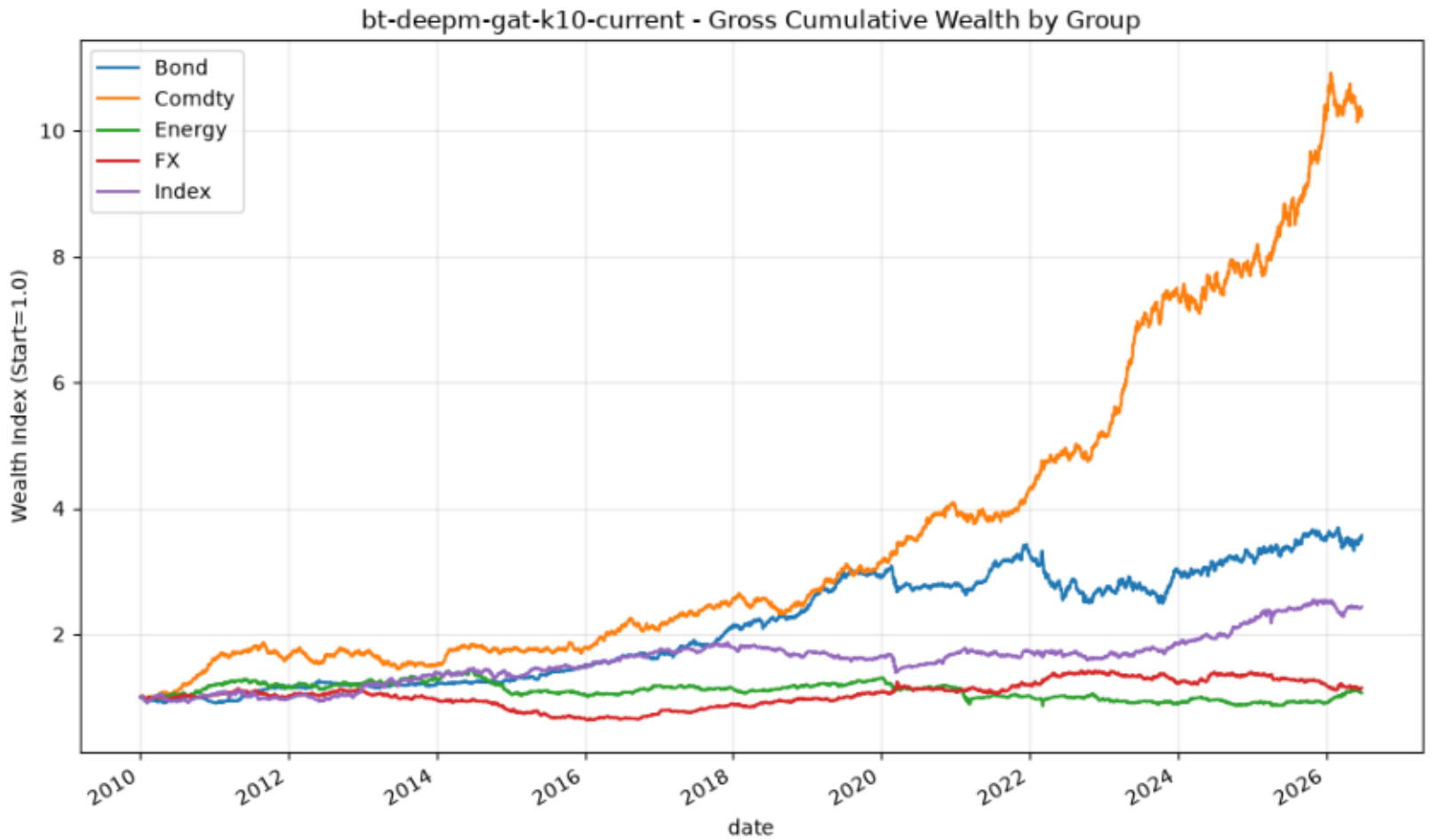
Net Drawdown



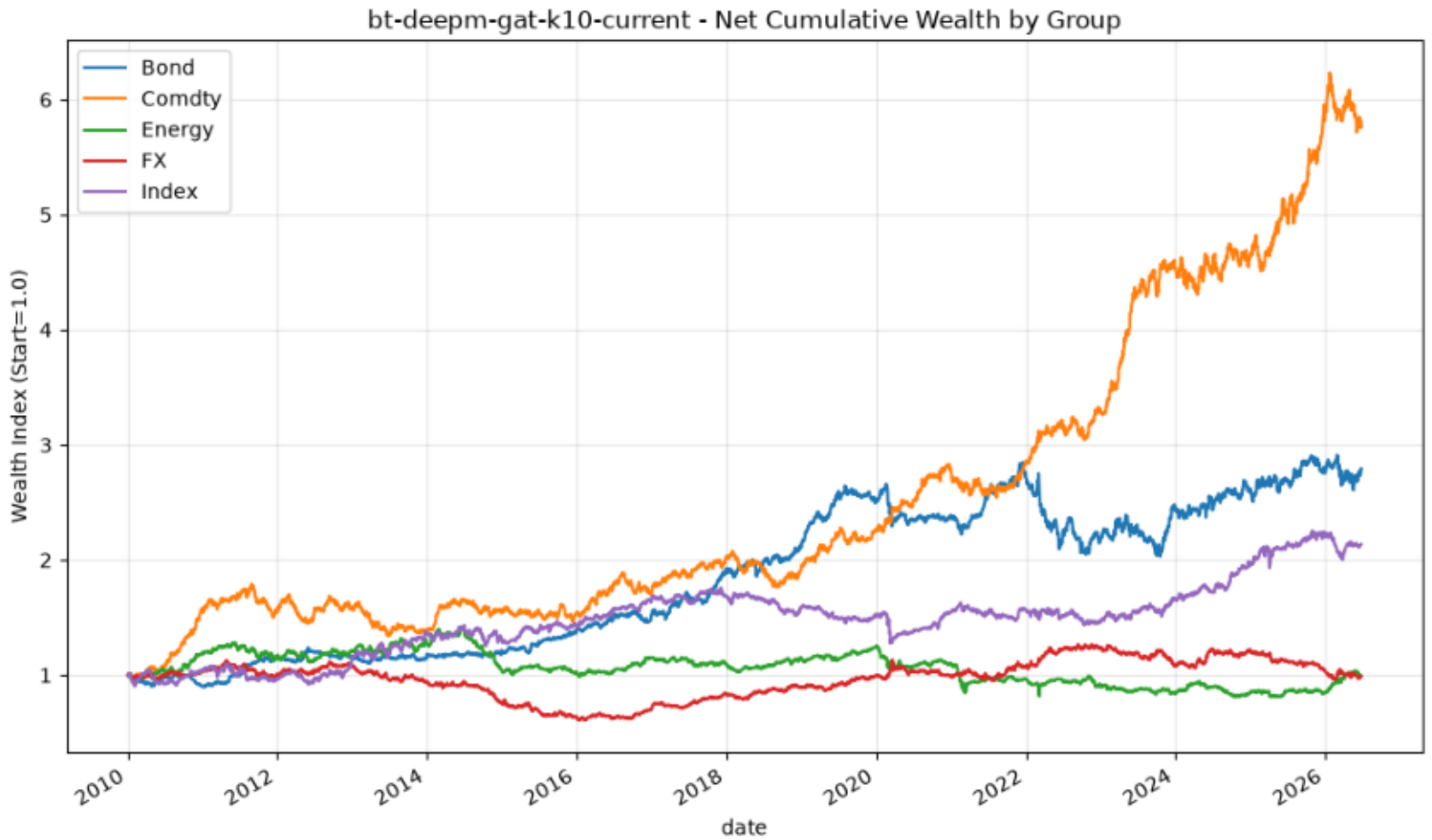
Annual Sharpe



Group Gross PnL



Group Net PnL



Interpretation Checklist

Capability: compare net Sharpe, Calmar, drawdown, turnover, and passive benchmark metrics.

Robustness: inspect annual Sharpe and drawdown pages for concentrated regime dependence.

Diversification: compare group gross/net PnL pages for asset-class concentration.

Selection quality: review whether top validation seeds also show stable test metrics.

Costs: use net-vs-gross spread and breakeven cost metrics to judge transaction-cost sensitivity.